Defect-Aware Fourier Bounds and Contact Budgets for Planar Circle Packing

A scoped Ram analysis of a sharp certificate

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Abstract

We study the physical realization, infinite interpolation, and continuum sign obligations in the September 23, 2026 manuscript *A sharp Fourier certificate for planar circle packing*. Its sharp-witness theorem is a source result, not a consequence of finite matrix invertibility alone. We prove a defect-tolerant density theorem for actual Schwartz functions using a positive Fourier repair and an explicit cumulative packing estimate. For the source's normalized witness we derive directed contact budgets with a linear penalty at the first shell and quadratic penalties at higher squared-radius nodes. We also give finite strict-slack obstructions to exact sharpness and a coherent Bernstein error certificate that retains shared coefficient uncertainty. Original Ram finite-jet inversion and Bernstein derivations are compared on their stated domains. The original public interval verifier was run unchanged at its pinned version and passed; independent exact local checks supplement, but do not replace, the proofs. Contribution assessment: C3/5, source-specific quantitative density and contact-error bounds are proved, without a new global packing theorem or a claim that the underlying classical mechanisms are novel.

1 Scope, conventions, and status

We use

$$\widehat{u}(\xi) = \int_{\mathbb{R}^2} u(x) e^{-2\pi i x\cdot\xi}\,\mathrm{d}x, \qquad b = \frac{\sqrt{3}}{2}, \quad h = \frac{2}{5}, \quad \rho_* = \frac{1}{b}.$$

A packing is a locally finite set of centers whose distinct points have distance at least one. Its disks have radius $1/2$. For a periodic set $\mathcal{C} = \bigcup_{j=1}^{N}(L + t_j)$, the cosets are distinct, L is a full-rank lattice, $V = \mathrm{covol}(L)$, and its number density is $\rho = N/V$. Covered-area density is $\pi\rho/4$. For arbitrary packings we use upper densities over growing Euclidean balls; the density theorem below is uniform in the observation center. This convention differs from the unit-ball center density used in some references by a factor four.

The source [1] claims a real radial Schwartz function f with

$$f(0) = \frac{1}{b}, \qquad \widehat{f}(0) = 1, \qquad f(x) \le 0 \ (|x| \ge 1), \qquad \widehat{f}(\xi) \ge 0 \ (\xi \in \mathbb{R}^2). \tag{1.1}$$

The Cohn–Elkies theorem then gives $\pi/(2\sqrt{3})$, the classical planar packing density. The source's claimed advance is sharpness of this particular Fourier bound, not the first proof of planar packing

optimality. Our conditional uses of (1.1) are explicitly named. Neither a finite inverse nor approximate roots furnish these hypotheses.

Fresh public searches at the October 8 review date found the released source and the original Cohn–Elkies paper [2]. The latter supplies the classical positive-sum comparison and the sharpness question in its Conjecture 7.3. We read its admissibility definition, main theorem and proof, and the relevant conjecture statements, not its entire proof corpus. Sardari's Theorem 1.7 [3] gives infinitely many linearly independent radial Schwartz functions with simultaneous double zeros at triangular-lattice shells. Its statement does not supply the required global signs. The source's uniqueness is inside its prescribed cardinal family, not among all radial Schwartz functions.

The 2026 HSE thesis record [5] describes a modular-form construction of a planar magic function. Only its institutional summary was available in the inspected record; we use no theorem from an unread thesis. Böröczky–Radchenko–Ramos [4], whose abstract was freshly inspected, already use magic-function deficits for quantitative stability in dimensions eight and twenty-four. No historical novelty is claimed for extracting positive penalties from a Poisson defect. Our contribution is the explicit planar specialization, different first-shell order, and certified-error interface below, not their much stronger geometric stability conclusions.

2 The source's analytic architecture

2.1 Cardinal data and physical realization

In the source's initial coordinates $s = b|x|^2$ and $B = b^{-2} = 4/3$, put

$$A = \{0, 1, 3, 4, 7, 9\}, \quad \mathcal{N} = (12\mathbb{Z} + A) \cap (0, \infty), \quad P(s) = \prod_{a \in A} \left(2 \sin \frac{\pi(s-a)}{12}\right)^2.$$

For a real list $(c_n, d_n)_{n \in \{0\} \cup \mathcal{N}} \in \ell^1$ and a real constant C, the source defines the entire function

$$p(s) = P(s) \left[C + \sum_n \left\{ \frac{c_n}{(s-n)^2} + \frac{d_n}{s-n} \right\} \right]. \tag{2.1}$$

The apparent poles are removable. More importantly, its proof realizes $p(s) = \int_{-1}^{1} e^{i\pi t s} \, \mathrm{d}\mu(t)$ with a finite complex measure and conjugate symmetry. A uniform total-variation bound permits the infinite list, differentiation on the real axis, and Gaussian multiplication. Thus $q_p(x) = e^{-\pi h s} p(s)$ is a real radial Schwartz function. The Fourier kernel is

$$\lambda(t) = \frac{i}{b(t+ih)}, \quad z(t) = -\frac{B}{t+ih} - ih, \quad K_p(s) = \int \lambda(t) e^{i\pi z(t) s} \, \mathrm{d}\mu(t), \quad \widehat{q_p}(\xi) = e^{-\pi h s} K_p(s). \tag{2.2}$$

For real $|t| \le 1$, $\operatorname{Im} z(t) \ge 26/435 > 0$. The Gaussian identity with positive real part justifies this Fourier transform. It is not a formal transform of a finite feature vector.

Two lists give $H_1 = p_1 + K_{p_2}$ and $H_2 = p_2 + K_{p_1}$, hence an exact Fourier pair $F = e^{-\pi h s} H_1$, $\widehat{F} = e^{-\pi h s} H_2$. The fixed zero-node data are $(c_{1,0}, d_{1,0}, C_1) = (1, .44, -.013)$ and $(c_{2,0}, d_{2,0}, C_2) = (0, -.368, .017)$. All printed terminating decimals here and below denote exact rationals. At every positive node, the desired equations are

$$H_1(n) = H_2(n) = H_2'(n) = 0, \qquad H_1'(n) = -Q_n \mathbf{1}_{\{n=1\}},$$

where $Q_n > 0$ is the leading coefficient of $P(n+u)$.

2.2 Infinite inverse, finite arithmetic, and continuum signs

The source's operator S acts on the *unweighted* $X = \ell^1(\mathcal{N}; \mathbb{R}^2)$, with all c coordinates preceding the d coordinates in each finite block. Its coupled equations are $x_1 - Sx_2 = g_1$, $x_2 - Sx_1 = g_2$. The uniform output-tail estimate is over *all input columns*; this is what makes S a bounded operator and its finite-row truncations operator-norm approximations. A small matrix alone is insufficient.

The first Schur elimination uses $J = \{1, 3, 4\}$ and $E = \mathcal{N} \cap [7, 100]$. Bounds for both $I - S$ and $I + S$ on J, cross terms, and the remaining rows yield the finite inverse bound 90. A second elimination uses the full output tail beyond 100, $\delta = 5 \cdot 10^{-10}$, and $\|S\| < 28$. The tail Schur defect is at most $2521\delta < 1$; the resulting resolvent estimate is

$$\|x\| \le 91 \, \|r_{\text{finite}}\| + 2540 \, \|r_{\text{tail}}\| \, .$$

The tabulated lists therefore lie within $3 \cdot 10^{-5}$ of the exact lists in the source's ℓ^1 norm. This conclusion uses both analytic tail bounds and finite arithmetic, not continuity of an unspecified limit of matrices.

For signs, 84 oriented half-gaps and degree-28 Bernstein polynomials cover $[0, 41.5]$ on the Fourier side and $[1, 41.5]$ on the physical side. Before dividing by u^ν, the source deletes the approximate lower jets; the exact function has those jets zero. A Taylor remainder controls the difference between exact and tabulated quotients. In the unbounded region, an exact double-zero argument, a second-derivative bound, and $P(s) \ge .68 \operatorname{dist}(s, \mathcal{N})^2$ supply the sign, not extrapolated samples. The source's numerical margins will be retained as imported constants where used below.

2.3 Normalization is an exact identity

Write $G_i(s) = e^{-\pi h s} H_i(s)$. The covolume-one triangular lattice has $s_a \in \mathcal{N}$ for every nonzero vector and exactly six vectors at $s_a = 1$. Its dual is a rotation. Dilation and Poisson summation give

$$\sum_a G_1(t s_a) = t^{-1} \sum_a G_2(s_a / t).$$

The Schwartz majorants justify differentiation locally uniformly in $t > 0$. Evaluation and first differentiation at $t = 1$ give

$$F_0 := F(0) = \widehat{F}(0) = 6 Q_1 e^{-\pi h} > 0, \qquad f(x) = \frac{F(x/\sqrt{b})}{b F_0}, \quad \widehat{f}(\xi) = \frac{\widehat{F}(\sqrt{b}\,\xi)}{F_0}. \tag{2.3}$$

In particular

$$\begin{aligned} f_{\text{rad}}(r) &= \frac{e^{-\pi h r^2} H_1(r^2)}{b F_0}, \qquad \widehat{f}_{\text{rad}}(r) = \frac{e^{-\pi h b^2 r^2} H_2(b^2 r^2)}{F_0}, \\ f'_{\text{rad}}(1) &= -\frac{1}{3b} = -\frac{2}{3\sqrt{3}}. \end{aligned} \tag{2.4}$$

The slope is with respect to physical radius, not squared radius. The second differentiated identity $\sum_n r_\Lambda(n) n^2 (G_1''(n) - G_2''(n)) = 2 G_2(0)$ is already in the source and is not counted as a new result.

3 A density bound allowing certified sign defects

The next results do not assume the sharp source witness exists. They start with an actual function on $\mathbb{R}^2$.

Lemma 3.1 (Cumulative leakage bound)**.** *Let $\mathcal{C}$ be any one-separated set containing x. Let $E : [1,\infty) \to [0,\infty)$ be continuously differentiable, nonincreasing, and satisfy $t^2E(t) \to 0$ and $\int_1^\infty tE(t)\,\mathrm{d}t < \infty$. Then*

$$\sum_{y\in\mathcal{C}\setminus\{x\}} E(|x-y|) \le \mathcal{K}(E) := 8E(1) + \int_1^\infty (8t+4)E(t)\,\mathrm{d}t. \tag{3.1}$$

For $E(t) = e^{-\pi\sigma t^2}$, $\sigma > 0$, a convenient upper bound is

$$\mathcal{K}(E) \le L(\sigma) := \left(8 + \frac{6}{\pi\sigma}\right) e^{-\pi\sigma}. \tag{3.2}$$

Proof. Translate x to zero. If $n(t)$ counts other centers of radius at most t, their disjoint open disks of radius $1/2$, together with the disk at zero, fit in $B(0, t+1/2)$. Comparing areas gives $n(t) \le 4(t+1/2)^2 - 1 = 4t^2 + 4t$ for $t \ge 1$. Since $E(r) = \int_r^\infty [-E'(t)]\,\mathrm{d}t$, Tonelli's theorem gives the sum as $\int_1^\infty n(t)[-E'(t)]\,\mathrm{d}t$; points at radius one are included. Integration by parts of the majorant proves (3.1), with boundary value $8E(1)$. For the Gaussian, $\int_1^\infty tE(t)\,\mathrm{d}t = E(1)/(2\pi\sigma)$ and $\int_1^\infty E(t)\,\mathrm{d}t \le \int_1^\infty tE(t)\,\mathrm{d}t$. This proves (3.2). □

Theorem 3.2 (Repaired Fourier certificate)**.** *Let $u \in \mathcal{S}(\mathbb{R}^2)$ be real and even. Suppose $a > 0$, $\epsilon \ge 0$, $\widehat{u}(0) + \epsilon > 0$, and*

$$\widehat{u}(\xi) \ge -\epsilon e^{-\pi a|\xi|^2} \quad (\xi \in \mathbb{R}^2).$$

Suppose $u(x) \le E(|x|)$ for all $|x| \ge 1$, where E satisfies Lemma 3.1. Every packing has upper number density at most

$$D(u, E, \epsilon, a) = \frac{u(0) + \mathcal{K}(E) + \epsilon a^{-1}[1 + L(1/a)]}{\widehat{u}(0) + \epsilon}. \tag{3.3}$$

Its covered-area upper density is at most $\pi D/4$. In particular, if $u(x) \le \delta e^{-\pi\sigma|x|^2}$ on $|x| \ge 1$, replace $\mathcal{K}(E)$ by $\delta L(\sigma)$, with $\delta \ge 0$, $\sigma > 0$.

Proof. The exact two-dimensional Fourier transform of $g_a(x) = a^{-1}e^{-\pi|x|^2/a}$ is $e^{-\pi a|\xi|^2}$. Thus $U = u + \epsilon g_a$ is Schwartz and $\widehat{U} \ge 0$. For a periodic packing, absolute convergence and Poisson summation give

$$\frac{N^2}{V}(\widehat{u}(0) + \epsilon) \le \sum_{j,k=1}^{N} \sum_{v\in L} U(v + t_j - t_k) \le N\left[u(0) + \frac{\epsilon}{a} + \mathcal{K}(E) + \frac{\epsilon}{a}L(1/a)\right].$$

The diagonal $v = 0, j = k$ is separated before applying Lemma 3.1 to each center. Division proves (3.3) for periodic packings.

For completeness the extension does not require a sign theorem for U. Take any finite portion of a packing in a half-open square of side T. Repeat it with periods $(T+1)\mathbb{Z}^2$; cross-copy distances are at least one. The periodic conclusion bounds the original square's count by $D(T+1)^2$. Tiling the plane by squares of side T, the number meeting any ball of radius R is at most $\pi R^2/T^2 + O_T(R+1)$, uniformly in its center. Consequently the upper number density is at most $D(1+1/T)^2$. Let $T \to \infty$. Disjoint radius-$1/2$ disks have covered area in a ball at most $\pi/4$ times the count in the ball of radius $R + 1/2$, proving the final assertion. □

Remark 3.3. The Fourier lower bound is global and requires its own unbounded-tail proof. A small residual at finitely many nodes does not supply it. The repair does not preserve interpolation zeros or the sharp ratio. It produces a valid, generally weaker density bound for an imperfect function. The theorem is useful even when the available certificate is insufficient to prove either exact global sign.

3.1 A worked certified-error budget

For $a = \sigma = 1$, $\pi > 3$ and $e^3 > 20$ imply $L(1) < 1/2$. The latter inequality follows already by summing the positive exponential series through degree eight. Hence the sufficient rationally organized bound is

$$\rho \le \frac{u(0) + \delta/2 + 3\epsilon/2}{\widehat{u}(0) + \epsilon}. \tag{3.4}$$

For example, *if* a function satisfies the global assumptions, $u(0) = \rho_*$, $\widehat{u}(0) = 1$, and $\delta = \epsilon = 10^{-6}$, then $\rho \le (\rho_* + 2 \cdot 10^{-6})/(1 + 10^{-6})$. This is a conditional worked error budget, not an assertion that the rounded source table has these global Gaussian defects. In particular, the slower of the physical and transformed decay rates must be accounted for before assigning σ or a.

4 Periodic defects and shell-sensitive stability

This section assumes the exact source witness (1.1), including its physical realization, normalization, and sign certificates. All sums below converge absolutely. An off-diagonal index is a triple (j, k, v) with $1 \le j, k \le N$, $v \in L$, other than $(j, j, 0)$. Write $r_{jkv} = |v + t_j - t_k|$ and $A(w) = \sum_j e^{2\pi i w \cdot t_j}$.

Proposition 4.1 (Exact positive defect ledger)**.** *For every periodic packing,*

$$\rho_* - \rho = \frac{1}{N} \sum_{\text{off}} [-f_{\text{rad}}(r_{jkv})] + \frac{1}{NV} \sum_{w \in L^* \setminus \{0\}} \widehat{f}(w) |A(w)|^2. \tag{4.1}$$

Both terms are nonnegative. Thus any selection of distinct indices in either sum has total normalized penalty at most $\Delta := \rho_* - \rho$.

Proof. Poisson summation writes the common energy as $Nf(0) - \sum_{\text{off}}[-f]$ and as $N^2/V + V^{-1} \sum_{w \ne 0} \widehat{f}(w)|A(w)|^2$. Equating and dividing by N gives the result. Separation supplies the physical sign, and the Fourier sign is global. □

This is the quantitative form of a classical equality argument. The weights, range constants, and distinction of root orders are the useful source-specific information. We count *directed* neighbors per center. An undirected edge count is half the directed count when both orientations are included; an extra factor two must not be silently inserted.

Lemma 4.2 (Explicit source shell penalties)**.** *Retain the source's quotient and tail certificates. For every $S \ge 1$ and $1 \le r \le \sqrt{S}$,*

$$-f_{\text{rad}}(r) \ge c_S \operatorname{dist}(r^2, \mathcal{N})^2, \qquad c_S = \frac{.00448\, e^{-\pi h S}}{bF_0}. \tag{4.2}$$

On the whole first half-gap $1 \le r \le \sqrt{2}$,

$$-f_{\text{rad}}(r) \ge c_{\text{lin}}(r - 1), \qquad c_{\text{lin}} = \frac{.336\, e^{-2\pi h}}{bF_0}. \tag{4.3}$$

Proof. On $1 \le s \le 2$ the source's exact first quotient is at least .168, so $-H_1(s) \ge .168(s - 1)$. Here $\operatorname{dist}(s, \mathcal{N}) = s - 1 \le 1$, including the tie at two. On the other finite half-gaps, the quotient at the nearest node is at least .006, yielding $-H_1(s) \ge .006 \operatorname{dist}(s, \mathcal{N})^2$. For $s \ge 41.5$ the source's proof gives $-H_1(s) \ge (.011 \cdot .68 - .003) \operatorname{dist}(s, \mathcal{N})^2 = .00448 \operatorname{dist}(s, \mathcal{N})^2$. The nearest-node statement includes midpoint ties; the polynomial zeros extend periodically and the nearest positive node exists on this range. Multiplying by $e^{-\pi h s}/(bF_0)$ proves (4.2). On the first interval use $s = r^2 \le 2$ and $r^2 - 1 = (r - 1)(r + 1) \ge 2(r - 1)$ to obtain (4.3). □

Theorem 4.3 (Quantitative contact budgets)**.** *Let $S \geq 1$ and $\tau > 0$. The average number of directed neighbors per center with $r \leq \sqrt{S}$ and $\operatorname{dist}(r^2, \mathcal{N}) \geq \tau$ is at most*

$$\frac{\Delta}{c_S \tau^2}. \tag{4.4}$$

For $0 < \eta \leq \sqrt{2} - 1$, the average number with $1 + \eta \leq r \leq \sqrt{2}$ is at most

$$\frac{\Delta}{c_{\mathrm{lin}} \eta}. \tag{4.5}$$

For any finite $W \subset L^* \setminus \{0\}$ such that $\widehat{f}(w) \geq \gamma > 0$ on W,

$$\frac{1}{NV} \sum_{w \in W} |A(w)|^2 \leq \frac{\Delta}{\gamma}. \tag{4.6}$$

Proof. Each selected physical summand is at least $c_S \tau^2$ or $c_{\mathrm{lin}} \eta$, respectively. Their sums are bounded by $N\Delta$ in (4.1). The same argument on the dual sum gives (4.6). These are finite-window counts even though the full identity has infinitely many nonnegative terms. □

The budgets can be added for *disjoint* physical index sets, and then combined with the dual penalty in (4.1). Overlapping physical selections must not charge the same defect twice. The constants deteriorate with S, as expected from Schwartz decay. They do not force all long-range distances close to nodes, fix a contact graph, control angles, or identify an isometry to a triangular packing. The dual lattice and its window may also depend on the packing period.

4.1 A first-shell certificate based directly on curvature

Proposition 4.4. *Suppose $f_{\mathrm{rad}}(1) = 0$, $f'_{\mathrm{rad}}(1) = -a_0$, $a_0 = 1/(3b)$, and a certified bound $|f''_{\mathrm{rad}}| \leq M$ holds on $[1, 1 + d_0]$, $M \geq 0$, $d_0 > 0$. Then for $0 \leq t \leq d_0$,*

$$-f_{\mathrm{rad}}(1 + t) \geq a_0 t - \tfrac{1}{2} M t^2.$$

Consequently the linear contact-count constant $a_0/2$ is valid on $[1, 1 + d]$, where $d = \min(d_0, a_0/M)$ and $a_0/0$ means $+\infty$.

Proof. The integral Taylor remainder is bounded by $M \int_0^t (t - v)\, \mathrm{d}v = Mt^2/2$. The stated restriction on t gives the linear lower bound, and Proposition 4.1 then gives the count just as before. □

There is an explicit, though conservative, source-compatible M. Its exact lists have norm less than 2.3, and the folded density envelopes give each total variation less than 80: $2.3(56 + 48 + 35 + 25 + 19 + 6.284)/6 + .017(25) < 80$. For $s \geq 0$, (2.2) and $|\lambda| < 3$, $|z| < 3$ imply $|H_1^{(k)}(s)| \leq T_k := 80\pi^k + 240(3\pi)^k$. Set $A_1 = T_1 + \pi h T_0$ and $A_2 = T_2 + 2\pi h T_1 + (\pi h)^2 T_0$. The chain rule in (2.4) gives

$$|f''_{\mathrm{rad}}(r)| \leq M_0 := \frac{2A_1 + 8A_2}{bF_0} \qquad (1 \leq r \leq \sqrt{2}). \tag{4.7}$$

Thus $d_0 = \sqrt{2} - 1$ and $M = M_0$ are legitimate without an inferred numerical curvature. A tighter interval curvature bound can greatly enlarge this local range. The first-half-gap bound (4.3) instead offers a longer fixed interval with a different constant.

5 Strict residual obstructions to sharpness

Proposition 5.1 (Finite lower bounds on the sharpness gap)**.** *Let u be real, radial and Schwartz, $\widehat{u}(0) > 0$, $u(r) \le 0$ for $r \ge 1$, and $\widehat{u} \ge 0$ everywhere. Let $\Lambda_0 = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, b)$, of covolume b. Then*

$$u(0) - \frac{\widehat{u}(0)}{b} = \sum_{\lambda \in \Lambda_0 \setminus \{0\}} [-u(\lambda)] + \frac{1}{b} \sum_{\xi \in \Lambda_0^* \setminus \{0\}} \widehat{u}(\xi). \tag{5.1}$$

Let $m(n) = \#\{(j,k) \in \mathbb{Z}^2 : j^2 + jk + k^2 = n\}$. If I, J are finite sets of positive represented integers and $-u_{\mathrm{rad}}(\sqrt{n}) \ge d_n \ge 0$ for $n \in I$, $\widehat{u}_{\mathrm{rad}}(\sqrt{4n/3}) \ge e_n \ge 0$ for $n \in J$, then

$$\frac{u(0)}{\widehat{u}(0)} - \rho_* \ge \frac{\sum_{n \in I} m(n) d_n + b^{-1} \sum_{n \in J} m(n) e_n}{\widehat{u}(0)}. \tag{5.2}$$

Proof. Poisson summation on Λ_0 gives (5.1) after isolating both origins. The dual is a rotation of $b^{-1}\Lambda_0$, so its squared lengths are $4n/3$ with the same multiplicities. Every summand is nonnegative. Retain the selected shells and divide by the positive $\widehat{u}(0)$. □

For instance $m(1) = 6$, so one rigorous physical interval lower bound $-u_{\mathrm{rad}}(1) \ge 10^{-8}$ and $\widehat{u}(0) = 1$ forces a number-ratio gap at least $6 \cdot 10^{-8}$. A rigorous interval for a *positive* residual is enough to obstruct exact sharpness, provided the global admissibility and sign hypotheses already hold. An interval merely containing zero cannot certify a zero.

The distinction between lattice shells and the source's larger node set matters. The integer 15 belongs to $\mathcal{N}$ but is not represented by j^2+jk+k^2. Indeed, modulo five the equation $j^2+jk+k^2 = 0$ forces both variables to vanish: for $k \ne 0$ its quadratic discriminant is $-3 = 2$, a nonsquare modulo five. It would follow that 25 divides 15, a contradiction. A residual at that extra node alone is not a triangular-lattice slack term in (5.2). The additional zeros may still be needed by the source's interpolation scheme.

5.1 Why approximate jets do not certify a touching sign

At an interior zero, a nonnegative differentiable real function must have derivative zero. Near a double zero consider

$$q(u) = cu^2 + \beta u + \alpha, \qquad c > 0.$$

Completing the square shows that $q \ge 0$ on all of $\mathbb{R}$ if and only if $4c\alpha \ge \beta^2$. On a prescribed finite interval one instead tests the endpoints and the vertex when the vertex lies inside. In particular, if $\alpha = 0$ and $\beta \ne 0$, the sign fails on one side arbitrarily close to zero, however small $|\beta|$ is. Even a constant error $-\varepsilon$ defeats cu^2 at the center. At the physical boundary $r = 1$ only the right side is required, so the simple negative slope in (2.4) is allowed. These observations explain why the source's exact lower jets must be retained when converting its numerical coefficient table into a global sign proof. They do not reveal an error in that source procedure.

6 A coherent finite moment certificate

The original Ram clearance proof [7] derives the power-to-Bernstein identity, the convex-hull bound, and exact rational subdivision. We now retain the correlations of coefficient errors instead of rounding each derived monomial independently.

Proposition 6.1 (Shared-error Bernstein bound)**.** *On $v \in [0,1]$ suppose an actual real trace has the representation*

$$Q(v) = \widetilde{Q}(v) + \sum_{j=1}^{m} z_j q_j(v) + R(v), \qquad \sum_{j=1}^{m} w_j |z_j| \le \varepsilon, \quad |R(v)| \le \eta,$$

where $\widetilde{Q}, q_j$ are polynomials of degree at most d, $w_j > 0$, and $\varepsilon, \eta \ge 0$. Write their degree-d Bernstein coefficients as $\widetilde{\beta}_k, C_{kj}$, respectively. Then

$$Q(v) \ge \min_{0 \le k \le d} \left[\widetilde{\beta}_k - \varepsilon \max_{1 \le j \le m} \frac{|C_{kj}|}{w_j} \right] - \eta. \tag{6.1}$$

The row error radius is the exact support function of this weighted ℓ^1 error ball. It is not necessarily the exact minimum of Q. The same assertion holds on each exact de Casteljau subdivision, using the transformed rows C and baseline coefficients.

Proof. For the Bernstein basis $B_k^d(v) = \binom{d}{k} v^k (1-v)^{d-k}$, the identity $v^r = \sum_{k=r}^{d} [\binom{k}{r} / \binom{d}{r}] B_k^d(v)$ follows by the binomial theorem after canceling the binomial factors. Thus linear transformation of the complete polynomial gives row k equal to $\widetilde{\beta}_k + \sum_j C_{kj} z_j$. Its minimum over the error ball is $\widetilde{\beta}_k - \varepsilon \max_j |C_{kj}|/w_j$: the inequality is weighted ℓ^1–ℓ^∞ duality, and equality is obtained by putting all error mass on a maximizing coordinate with opposite sign. Bernstein weights are nonnegative and sum to one. Their convex combination, followed by subtraction of η, proves (6.1). Subdivision acts linearly on the entire polynomial and on every error column, so the proof applies without independent re-rounding of shared errors. □

6.1 An exact example with a strict improvement

Let $Q(v) = 1 + z(v - v^2)$, $|z| \le 1$. In degree two the error column is $(0, 1/2, 0)$, so (6.1) certifies $Q \ge 1/2$. If the coefficients of v and v^2 are instead treated as unrelated errors of radius one, the endpoint row gives the useless bound -1. Splitting at $1/2$ yields error rows $(0, 1/4, 1/4)$ and $(1/4, 1/4, 0)$. The coherent lower bound becomes $3/4$, attained at $v = 1/2, z = -1$. This is an exact rational certificate improvement for shared errors, not an empirical speedup claim.

For the source's quotient moments, the feature rows are linear in the cardinal lists. Orientation contributes the signed factor y^r on a left half-gap; it cannot be replaced by $|y|^r$. One may therefore assemble a coherent sensitivity row *before* applying its dual norm. The baseline must still be the lower-jet-deleted quotient used by the source, with genuine exact-zero justification. For an infinite list, one either proves a uniform bound over all error columns or splits off a finite head and supplies an independent analytic tail remainder η. We have not recomputed the source's 2436 margins using this alternative. Our executed source verification uses its original enclosure method.

7 Original Ram comparison and failure gates

Freshly inspected passages of the original finite-jet report [6] state $g_N = K_N f_N + r_N$ and $f_N = K_N^{-1} g_N - K_N^{-1} r_N$ only when the matrix is invertible. Polynomial action may be exact while the output fails to stay in the chosen polynomial space; compatible global jets and realization remain separate issues. Its infinite-order proposition first requires a specified Banach space, justified action of the physical kernel, and bounded operator inversion. These are precisely the distinctions needed here, although the source's S is not itself that finite triangular moment matrix.

The original counterexample is especially decisive. On $\ell^2(\mathbb{N}_0)$, let $(Ux)_j = -2x_{j+1}$. Every leading finite block of $I + U$ is triangular with determinant one, yet $x_j = 2^{-j}$ is a nonzero vector in its infinite kernel. The finite inverse's last-column norm grows with the truncation. Neither finite rank nor exact finite arithmetic eliminates this obstruction. M183 avoids the inference by proving complete output-tail operator bounds and two full Schur eliminations. We retain, rather than replace, that valid source argument.

The clearance original [7] also separates nominal polynomial feasibility from physical trajectory validity and from a nonpolynomial remainder. Its exact cubic path has clear endpoints but an interior collision. For M183 the analogous distinctions are: finite coefficient data versus an actual Schwartz Fourier pair; values at interpolation nodes versus signs on half-gaps; and compact half-gaps versus the whole unbounded radial tail. Proposition 6.1 improves an error representation, not any of those realization or global-domain hypotheses.

Object	What is certified	What is not implied
Finite inverse	Retained coordinates, with residual	Infinite bounded inverse
Bernstein hull	Entire stated polynomial interval	Unknown-function tail sign
Exact zero jets	Correct touching order at a node	Positivity between nodes
Defect repair	Density bound from global envelopes	Preservation of sharpness
Contact budget	Directed finite-window penalties	Triangular contact graph or crystallization

8 Source map and verification boundaries

8.1 Complete scientific-source coverage

All 3740 lines of the 16 cached text files were displayed in bounded literal reads: thirteen TeX files, bibliography, article README, and verification README. This is textual coverage, not a formal verification of every claim cited by those texts. The owned read ledger preserves each displayed interval, pre/post SHA-256 hashes, heading inventory, and the empty unread complement for these files.

Source file	Lines	Scientific dependency
introduction	193	Claim, density conventions, literature and zero-set scope
fourier	401	Cardinal realization, finite measure, Schwartz pair
interpolation	401	Complete ℓ^1 tails, both Schur eliminations
signs	437	Exact jets, oriented half-gaps, unbounded-tail barrier
normalization	311	Dilation Poisson, slope, periodic equality, curvature
certificates	572	Inputs, quadrature proof, matrices, residuals, signs
independent-certification	863	Alternative rational errors and Taylor enclosures
rational-check	173	Scope and proof of optional first-block check
main, three data files, figure	204	Assembly, exact tables and half-gap diagram
references and two READMEs	185	Bibliography and execution contract

The optional rational recipe is not the primary proof's numerical run. Its $N = 64$ program checks only the 6×6 first block and small inverse certificates. The rational $N = 256$ error theorem specifies how to enclose all finite moments but does not itself assert that all those rational sums were evaluated. Analytic continuation to complex quadrature disks is justified for the finite input lists; an arbitrary ℓ^1 frequency density is not thereby analytic on those disks. The signed half-gap

scaling, moment-error amplification in Bernstein rows, atom terms, and all-input-column tail scope are retained. No substantive correction to these inspected source chains was established.

8.2 Fresh original-private scope

The original report [6] was freshly read at lines 184–321 (notation, moment action, finite inverse and triangular hypotheses) and 1795–1875 (finite versus analytic infinite action, proof and counterexample, and adjacent qualifications). Its other 1728 lines were not read in this study. The entire 299-line original clearance chapter [7] was read, including its conversion proof, subdivision argument, remainder and model-realization cautions. The total fresh original-private coverage is 518 distinct lines. External chapter figures and that corpus's cited bibliographies were not independently inspected. No whole-corpus audit is claimed.

8.3 Original execution and independent tests

The local cache omitted the executable verification files. Fourteen public files were retrieved into an isolated owned build directory from commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; their Git blob hashes were checked against the pinned public listings. These comprise the verifier code, manifest, TSV inputs and the three corresponding TeX data tables. The primary runner, data loader, finite checker and scalar checker were literally read in full before execution. The optional rational runner was neither read in full nor executed. No cached source file or original verifier was modified.

The command ran with assertions enabled, Python's bytecode writes disabled, and the documented `python-flint==0.9.0` installed in the owned build directory. It exited zero in about thirty seconds; its newly generated summary says `pass`. It checked 51 finite nodes, both matrix signs, 102 residual pairs, 22 sign centers, 84 half-gaps (41 physical and 43 Fourier), 2436 individual Bernstein inequalities, and its stated scalar comparisons, including 22 named printed bounds. The 192-bit Arb run encloses moments with the source's proved 10^{-22} analytic quadrature allowance. This is a reproducible execution of the original calculation, not a new independent implementation or a machine proof of its analytic premises.

The separate author-written standard-library exact test suite passes 6589 cases comprising 44549 explicitly counted rational/integer assertions. Cases cover coherent support radii, exact subdivision, touching-root quadratics, Taylor budgets, finite packing counts and lattice residues, source rational constants, positive-ledger algebra, and finite versus infinite triangular inversion. A case is a stated parameter tuple or fixture; an assertion is one executed Boolean comparison. No floating-point agreement is counted as an exact assertion. Parent checks and the original checker are excluded from these totals. The finite tests do not establish infinite convergence, global positivity, or a new physical Fourier witness.

9 Advantages, costs, and further work

Useful increments. Theorem 3.2 gives a genuine fallback when exact sign certification is unavailable: a positive spectral repair has an explicit geometric leakage cost and applies to arbitrary packings, not only lattices or a fixed period. The cumulative count bound avoids assigning the full disk-area upper bound independently on every annulus. Theorem 4.3 converts the source's existing margins into explicit directed contact budgets and preserves the stronger linear first-shell behavior. Proposition 5.1 turns a rigorous strict residual into a quantitative obstruction to exact sharpness.

Proposition 6.1 demonstrates a strict rational enclosure gain when multiple derived coefficients share an error variable.

Costs and limitations. The leakage estimate sacrifices geometric information and is not expected to be optimal. A Gaussian Fourier repair may have expensive physical leakage; choosing a large a reduces its origin value but widens its physical support. Choosing a requires valid global Fourier envelopes, not optimization on sampled radii. The shell constant c_S decays exponentially with the window size, and a coarse curvature bound yields a very small first-shell Taylor window. Coherent rows cost storage or regeneration of error columns and may still require subdivision. None of these devices independently settles the exact sharp-witness construction or certifies a contact graph.

Feasible extensions. One can replace the Gaussian repair by another actual Schwartz pair with a proved nonnegative Fourier transform and an integrable physical majorant, using the same proof. A sum of Gaussian repairs can separate several spectral-defect scales; the corresponding physical leakage costs add by positivity. This does not justify choosing unrelated physical and Fourier arrays. On a fixed radial window, interval curvature and coherent sensitivity rows could improve the explicit constants without altering the source's exact-zero requirements. For an approximate witness, one should first certify all exterior envelopes and then report the density loss rather than silently substitute approximate interpolation identities into Poisson normalization.

A geometric near-crystallization result would additionally need control of missing neighbors, angular relations and compatible local patches. Small penalty on the source's enlarged node set does not by itself distinguish every triangular shell from extra nodes such as 15. Uniform passage from a period-dependent dual window to a period-independent geometric conclusion is another separate problem. These are research obligations, not conclusions of the present finite contact ledger.

10 Conclusion

The source combines actual Schwartz realization, complete infinite interpolation, finite enclosure arithmetic and continuum signs. Its valid separations must survive any Ram reformulation. The local results here preserve them while adding a quantitative sign-defect density theorem, shell-sensitive stability budgets and strict-sharpness diagnostics. Classical Poisson positivity, Taylor theory, area counting and Bernstein convexity receive no novelty credit. C3 records the demonstrated source-specific quantitative increment; applicability L3 is a different assessment. No new global packing theorem, universal optimality theorem, or unrestricted uniqueness of a Fourier witness is asserted.

References

[1] OpenAI, *A sharp Fourier certificate for planar circle packing*, September 23, 2026. Pinned public source and verification files, OpenAI Math repository.

[2] H. Cohn and N. Elkies, *New upper bounds on sphere packings I*, Annals of Mathematics **157** (2003), 689–714. doi:10.4007/annals.2003.157.689.

[3] N. Talebizadeh Sardari, *Higher Fourier interpolation on the plane*, arXiv:2102.08753v2 (2021), Theorem 1.7 (statement inspected).

[4] K. J. Böröczky, D. Radchenko and J. P. G. Ramos, *A quantitative stability result for the sphere packing problem in dimensions 8 and 24*, arXiv:2303.07908v2 (2024). Primary abstract inspected; no theorem from its unread proof is used.

[5] D. Zhitniaia, *Linear Programming Bound for Sphere Packing in Dimension 2*, HSE bachelor's thesis institutional summary (2026). Primary institutional record. Full thesis not inspected.

[6] M. Subbarao, original private Ram finite-jet and infinite-order kernel-symbol report, local file `TR-rev-1-old-tr147.tex`, passages specified in Section 8. Confidential local reference; not uploaded.

[7] M. Subbarao, original Ram Calculus volume 16 clearance chapter, local file `ch05d-clearance.tex`, especially Bernstein conversion, hull, subdivision and model-remainder sections. Confidential local reference; not uploaded.